

AI-Powered Customer Support Ticketing System

A Salesforce-Based Application for Automated Support Case Management and AI-Assisted Resolution

Project Team

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1. Project Description

The AI-Powered Customer Support Ticketing System is a Salesforce-based application designed to streamline and automate customer support operations for organizations that handle a high volume of customer issues. Built using core Salesforce Administrator capabilities together with Agentforce AI features, the system enables support teams to create, track, prioritize, and resolve tickets through a structured, rule-driven workflow rather than relying on ad-hoc manual handling.

The application is centered around the standard Case object, extended with custom supporting objects to capture additional ticket metadata, SLA (Service Level Agreement) tracking details, and categorization data that the standard Case object does not natively provide. When a customer support request enters the system, it is automatically evaluated against a defined set of criteria — combining declarative rule-based logic (Flows and Validation Rules) with Agentforce AI-assisted classification — to determine its category, priority, and the most appropriate queue or agent for resolution.

Support agents interact with the system through a customized Case management interface that surfaces AI-assisted suggestions, relevant knowledge references, and SLA countdown indicators, allowing them to act quickly and consistently. Administrators and support managers are provided with role-based access control, ensuring that sensitive case data and configuration settings are only accessible to authorized personnel, and are given visibility into team performance through a suite of reports and dashboards covering case volume, resolution time, SLA compliance, and agent productivity.

The project was implemented within a Salesforce Developer Org, using declarative development wherever possible — Flows, Validation Rules, Approval Processes, Permission Sets, and Reports — supplemented by Agentforce AI capabilities for intelligent case classification and agent assistance. This approach was deliberately chosen to keep the solution simple, maintainable, and reproducible without requiring custom Apex code or external AI service integration for its core functionality.

2. Objectives

The primary objectives of the AI-Powered Customer Support Ticketing System project are outlined below:

- To design a centralized Salesforce-based system for logging, tracking, and managing customer support tickets from creation through resolution.
- To reduce manual effort in case classification by automatically categorizing incoming tickets based on their subject, description, and origin.
- To ensure timely resolution of support requests through automated SLA tracking and escalation mechanisms.
- To improve agent productivity by surfacing AI-assisted suggested actions and relevant information at the point of case handling.
- To implement role-based access control so that case data, configuration, and reports are visible only to appropriately permissioned users.

- To provide support managers and administrators with visibility into support team performance through Salesforce Reports and Dashboards.
- To build the solution using core declarative Salesforce Admin features — Flows, Validation Rules, Permission Sets, and Agentforce AI — to keep the implementation simple, reliable, and easy to maintain.
- To design the system with scalability in mind, so that it can accommodate a growing volume of support tickets and additional case categories over time.

3. Problem Statement

Customer support teams are routinely responsible for triaging large volumes of incoming issues, each varying in urgency, complexity, and subject matter. In many organizations, this triage process is performed manually: a support agent or team lead reads each incoming request, determines its category and priority, and assigns it to the appropriate queue or individual agent.

This manual approach introduces several operational challenges. As ticket volume grows, manual classification becomes time-consuming and difficult to scale consistently. Different agents may categorize similar issues differently, leading to inconsistent prioritization and, in some cases, delayed responses to genuinely urgent issues. Without a structured SLA tracking mechanism, tickets can silently breach expected response or resolution timeframes, negatively impacting customer satisfaction.

Furthermore, agents handling a new ticket often lack immediate access to relevant historical context or suggested next steps, resulting in longer resolution times and inconsistent handling of similar issues. There is a clear need for a support ticketing system that automatically classifies and prioritizes incoming cases, actively tracks SLA compliance, and assists agents with relevant suggestions — while remaining transparent and auditable so that support managers can monitor and improve team performance over time.

In addition to these operational challenges, customer support organizations require a centralized platform that provides complete visibility into the lifecycle of every support request. Without a unified system, important information is often scattered across emails, spreadsheets, or multiple communication channels, making it difficult to monitor case progress and ensure accountability. This lack of centralized tracking can result in duplicated efforts, missed follow-ups, and limited visibility into overall support performance.

Another significant challenge is the absence of intelligent assistance during case resolution. Support agents frequently spend considerable time drafting responses, identifying appropriate troubleshooting steps, and determining the correct course of action for recurring issues. This repetitive effort reduces productivity and increases the likelihood of inconsistent customer communication. By integrating artificial intelligence into the support workflow, organizations can provide agents with context-aware reply suggestions and classification recommendations, enabling faster and more consistent service delivery while allowing agents to retain full control over the final response.

From a managerial perspective, the lack of real-time analytical insights makes it difficult to evaluate operational efficiency, identify bottlenecks, and measure compliance with service-level agreements. Managers require comprehensive reports and dashboards that provide visibility into case distribution, agent workload, resolution times, and SLA performance. Such insights support informed decision-making, resource allocation, and continuous process improvement.

Therefore, there is a need for an integrated, cloud-based customer support solution that combines Salesforce's declarative automation capabilities with Agentforce AI to automate repetitive tasks, improve case management, enhance agent productivity, and provide actionable business insights. The proposed AI-Powered Customer Support Ticketing System addresses these challenges by delivering a secure, scalable, and intelligent platform capable of supporting modern customer service operations while maintaining data integrity, transparency, and high service quality.

4. Existing System

In many existing support environments, customer issues are logged through email, phone calls, or basic web forms, and are then manually reviewed by a support agent or team lead to determine their category, urgency, and appropriate ownership. This process typically relies on the individual judgement of whoever first reviews the incoming request.

Some organizations use basic ticketing tools with limited automation, such as fixed routing rules based only on the channel through which a ticket arrived (for example, all email tickets going to a single shared queue). While this introduces a degree of structure, it does not account for the actual content, urgency, or category of the issue itself, and typically does not provide agents with any AI-assisted guidance or suggested resolution paths during case handling.

Most conventional customer support systems also provide limited visibility into Service Level Agreement (SLA) compliance. Agents are often required to manually monitor ticket deadlines and prioritize cases based on experience rather than automated reminders or predefined business rules. As a result, high-priority issues may remain unresolved for extended periods, increasing the likelihood of SLA breaches and negatively affecting customer satisfaction and organizational reputation.

Another limitation of traditional support environments is the lack of intelligent decision support. Agents typically rely on personal knowledge, previous experience, or internal documentation to determine appropriate responses for customer issues. This manual approach can lead to inconsistent resolutions, increased handling times, and variations in service quality, particularly when new or less experienced agents are involved.

Furthermore, many existing systems offer only basic reporting capabilities, requiring managers to manually compile performance metrics from multiple sources. Without comprehensive dashboards and real-time analytics, it becomes difficult to monitor agent productivity, identify recurring issues,

evaluate workload distribution, or measure overall support efficiency. This limits an organization's ability to make data-driven decisions and continuously improve customer service operations.

5. Limitations of Existing System

- Manual ticket triage is time-consuming and does not scale efficiently as support volume increases.
- Categorization and prioritization decisions vary between different agents, leading to inconsistent handling of similar issues.
- Basic routing rules based only on channel or source do not account for the actual content or urgency of a ticket.
- Existing systems often lack automated SLA tracking, making it difficult to detect and prevent response or resolution delays.
- Agents typically do not receive AI-assisted suggestions or contextual guidance while resolving a case, leading to longer resolution times.
- Limited reporting visibility makes it difficult for support managers to identify bottlenecks or measure team performance objectively.
- Access control is often coarse-grained, without clear role-based separation between agents, team leads, and administrators.

6. Proposed Solution

The AI-Powered Customer Support Ticketing System proposes a Salesforce-based alternative that replaces inconsistent manual triage with a structured, semi-automated workflow. The system is built around the standard Case object, extended with custom supporting objects and fields to capture ticket category, priority, SLA target timestamps, and AI-assisted classification metadata.

When a new case is created — whether through the Salesforce interface, a web-to-case form, or an email-to-case channel — a combination of Record-Triggered Flows and Agentforce AI-assisted classification evaluates the case subject and description to determine its likely category and priority. Validation Rules ensure that mandatory fields are captured accurately before a case can progress, and automated SLA timestamps are calculated based on the assigned priority level.

Rather than relying purely on rigid if-then rules, the system incorporates Agentforce AI capabilities to assist with case classification and to surface suggested actions or relevant knowledge references to the assigned agent, combining the reliability of declarative automation with the contextual assistance of AI. Email Alerts notify agents and support managers of new assignments, approaching SLA deadlines, and escalations, ensuring that no ticket is left unattended.

Support managers and administrators are granted access to Reports and Dashboards that summarize ticket volume, category distribution, SLA compliance rates, and individual agent performance, enabling data-driven decisions about staffing and process improvements.

7. Advantages

- Provides a centralized, structured system for logging and tracking every customer support ticket from creation to resolution.
- Reduces manual classification effort through a combination of rule-based Flows and Agentforce AI-assisted categorization.
- Automated SLA tracking and Email Alerts help ensure tickets are addressed and resolved within expected timeframes.
- AI-assisted suggested actions and relevant references help agents resolve cases more quickly and consistently.
- Role-based access control, implemented through Profiles and Permission Sets, ensures data is only visible to appropriately authorized users.
- Reports and Dashboards give support managers real-time visibility into team performance and SLA compliance.
- Built entirely using core declarative Salesforce Admin tools, making the solution easy to maintain, audit, and extend without heavy reliance on custom code.
- The modular design (custom objects, Flows, Validation Rules, Permission Sets) allows the system to be extended with additional ticket categories or business rules as requirements evolve.

8. Project Scenarios

Scenario 1: New Support Ticket Submission

A customer or support agent creates a new Case describing an issue — for example, a billing discrepancy or a technical product issue. Upon submission, a Record-Triggered Flow evaluates the case subject and description, working alongside Agentforce AI-assisted classification, to automatically assign a category and priority. SLA target timestamps are calculated based on the assigned priority, and the case is routed to the appropriate queue for agent action.

Scenario 2: Agent Handling an Assigned Case

A support agent opens an assigned case and is presented with AI-assisted suggested actions and relevant context alongside the standard case details. The agent updates the case status as they work through resolution, guided by the SLA countdown displayed on the case record, and the system sends automated Email Alerts if the case approaches its SLA deadline without resolution.

Scenario 3: Support Manager Reviewing Team Performance

A support manager accesses the Reports and Dashboards module to review key metrics such as total ticket volume by category, average resolution time, SLA compliance percentage, and individual agent performance. This visibility allows the manager to identify bottlenecks, rebalance workloads, and make informed staffing or process decisions.

9. Technical Architecture

The AI-Powered Customer Support Ticketing System follows a declarative, configuration-driven architecture native to the Salesforce Platform, separating data modeling, automation, AI-assisted classification, and presentation into distinct, maintainable layers.

9.1 Data Layer

The data layer is built around the standard Case object, extended with custom fields to capture category, priority, SLA target date/time, and resolution notes. Custom supporting objects are used to model additional ticket metadata — such as SLA policy definitions and category-specific attributes — that are not natively represented on the Case object, and are related to Case through lookup or master-detail relationships.

9.2 Automation Layer

The automation layer consists of Record-Triggered Flows that execute on Case creation and update, Validation Rules that enforce data quality before a record can be saved, and Email Alerts that notify relevant users of assignments, approaching SLA breaches, and escalations. This layer is responsible for calculating SLA timestamps, applying category and priority logic, and routing cases to the correct queue.

9.3 AI-Assistance Layer

Agentforce AI capabilities are layered on top of the automation layer to assist with case classification and to provide agents with suggested actions and contextual guidance while working a case, complementing the deterministic rule-based logic implemented through Flows and Validation Rules.

Figure 9.1 – Technical Architecture Diagram

Salesforce AI-Powered Customer Support Ticketing System

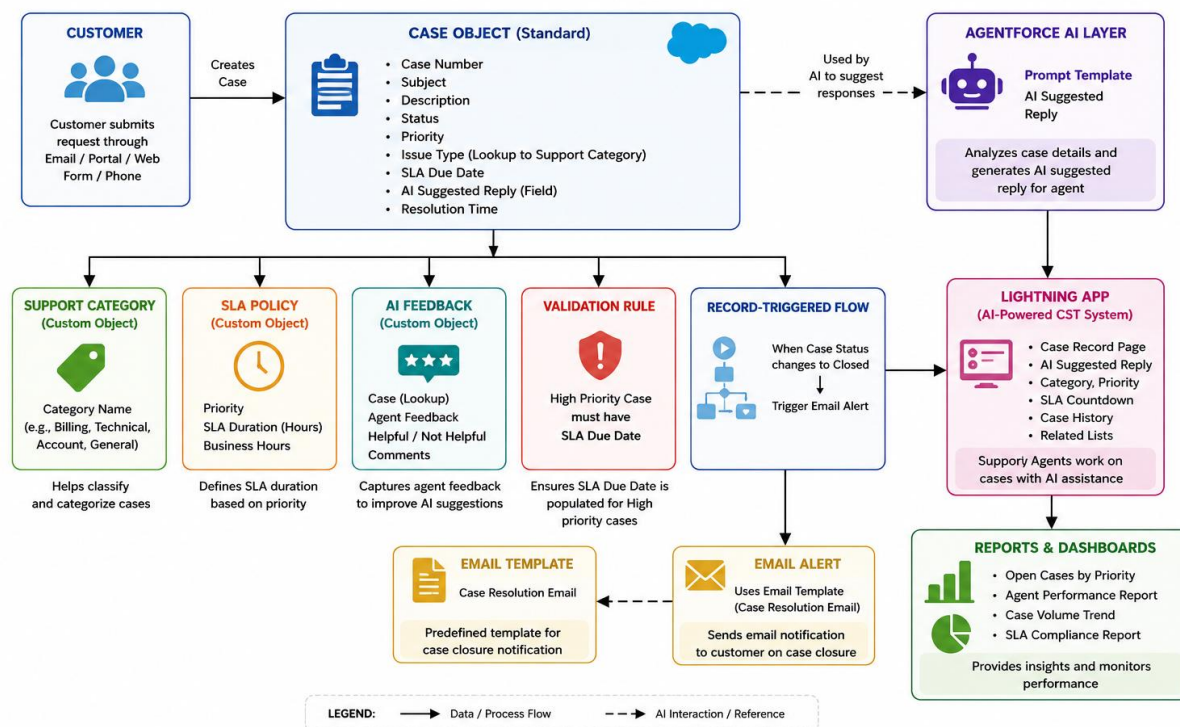


Figure 9.1: High-level technical architecture of the AI-Powered Customer Support Ticketing System

10. Data Flow Overview

The data flow begins when a new Case record is created, either directly by a support agent, through a Web-to-Case form, or via an Email-to-Case channel. Upon creation, a Record-Triggered Flow fires and evaluates the case's subject and description fields, working in conjunction with Agentforce AI-assisted classification, to determine the appropriate category and priority.

Validation Rules run at the point of save to ensure that mandatory fields — such as contact information, subject, and description — are populated before the case can progress. Once classified, the Flow calculates SLA target timestamps based on the assigned priority level and updates the relevant custom fields on the Case record. The case is then routed to the appropriate queue based on its category.

As an agent works the case, status changes and updates trigger additional Flow logic — for example, sending Email Alerts if the case is nearing its SLA deadline, or notifying a support manager if an escalation threshold is reached. All case data, including its full field history, is retained within Salesforce, feeding into the Reports and Dashboards layer used for ongoing performance monitoring.

After the case has been assigned, support agents access it through the AI-Powered CST System Lightning App, where all relevant customer information, case details, SLA deadlines, and AI-generated suggestions are available within a single interface. The Agentforce AI component analyzes the case context and generates a suggested customer response, allowing agents to review, edit, and send a professional reply while retaining full control over the final communication. This AI-assisted workflow helps reduce response times and ensures greater consistency in customer interactions.

Throughout the lifecycle of the case, every update—including changes to priority, ownership, status, and resolution details—is recorded within Salesforce, providing a complete audit trail for future reference and compliance purposes. Automated notifications and email alerts ensure that both customers and support personnel remain informed about significant case events, such as successful case resolution or status changes. These automation features minimize manual follow-up activities and improve overall communication efficiency.

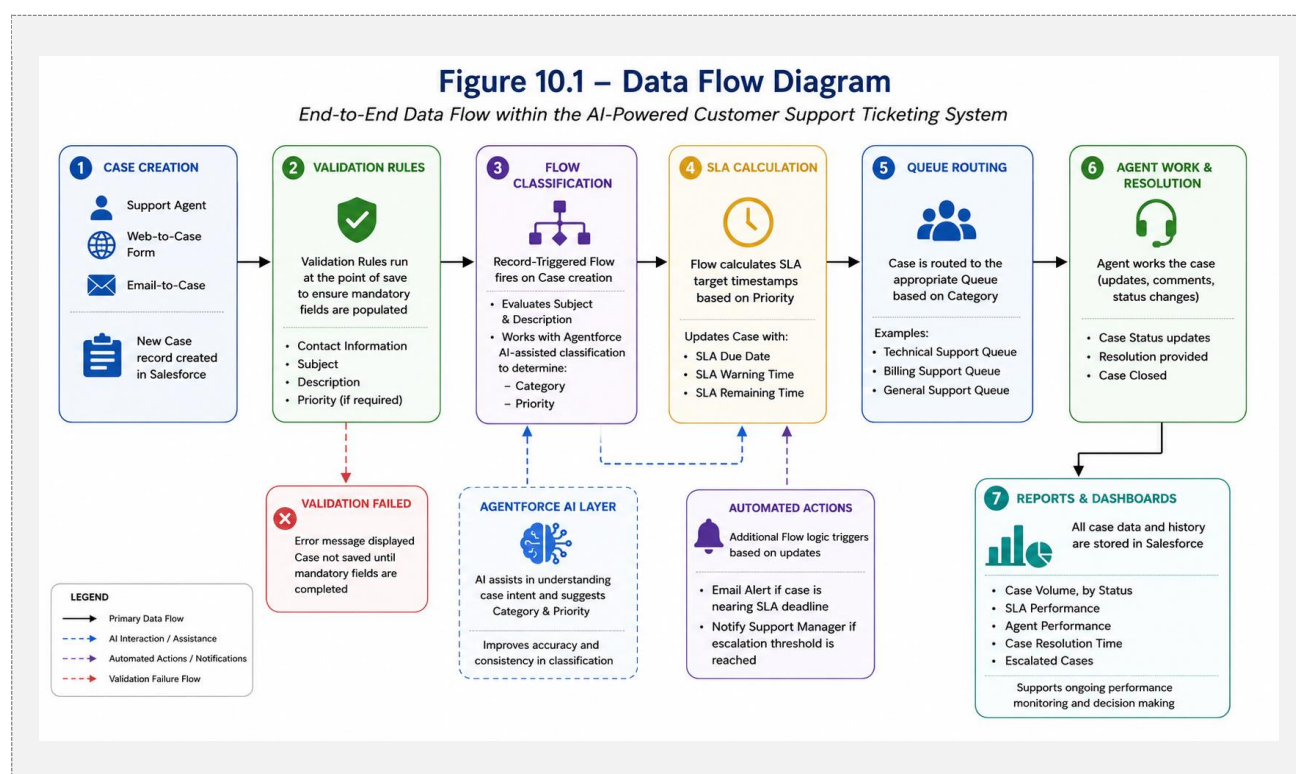


Figure 10.1: End-to-end data flow within the ticketing system

11. Project Workflow

The development of the AI-Powered Customer Support Ticketing System was carried out in a systematic manner, progressing from requirement gathering through Salesforce configuration to AI integration and testing. The workflow is organized into the following major milestones, each of which builds upon the outputs of the previous stage.

- Milestone 1 – Requirement Gathering: Understanding stakeholder needs and defining functional and non-functional requirements.
- Milestone 2 – Data Modeling: Designing the Case object extensions and custom supporting objects required to capture ticket metadata.
- Milestone 3 – Backend Development & Configuration: Building Validation Rules, Record-Triggered Flows, and Email Alerts for case classification, SLA tracking, and notifications.
- Milestone 4 – Agentforce AI Integration: Configuring AI-assisted case classification and suggested actions for support agents.
- Milestone 5 – UI/UX Development & Customization: Designing Lightning record pages, list views, and layouts to surface relevant information to agents and managers.
- Milestone 6 – Security, Reports & Testing: Configuring Profiles and Permission Sets, building Reports and Dashboards, and conducting functional testing across the system.

12. Execution Roadmap

The execution roadmap outlines the phased approach followed to implement the Salesforce AI-Powered Customer Support Ticketing System. The project is executed in a structured manner to ensure smooth development, testing, and deployment.

Phase 1: Requirement Analysis & Planning

Understand business requirements, identify stakeholders, define project scope, and finalize the system architecture and data model.

Phase 2: Environment Setup

Set up the Salesforce Developer Org, configure basic settings, enable Lightning Experience, and prepare standard and custom objects.

Phase 3: Data Model & Security Configuration

Create required objects and fields, establish relationships, and configure profiles, permission sets, and role-based access.

Phase 4: Automation & Case Classification

Implement record-triggered flows to automatically classify cases, assign priority, and calculate SLA due dates.

Phase 5: Agent Assistance & Feedback Loop

Configure suggested actions for agents, enable quick actions, and implement the AI feedback loop to capture agent responses.

Phase 6: Reporting & Monitoring

Create reports to track open cases, SLA breaches, and agent performance for management visibility.

Phase 7: Testing & Validation

Test all functionalities with different scenarios to ensure accuracy, automation reliability, and data integrity.

Phase 8: Deployment & Documentation

Activate all configurations, finalize documentation, and prepare the system for demonstration or handover.

13. Milestone 1: Requirement Gathering

The requirement gathering phase focused on understanding the challenges faced by customer support teams in handling support requests efficiently. Stakeholders required a centralized system to manage customer issues, reduce manual effort in case classification, ensure timely resolution through SLA tracking, and improve agent productivity.

The system needed to automatically categorize cases, assign appropriate priority, assist agents with suggested actions, and provide visibility into support performance through reports. Based on these discussions, functional and non-functional requirements were identified, emphasizing ease of use, automation, data accuracy, role-based access control, and scalability.

The solution was designed to be implemented using core Salesforce features in a Developer Org, combined with Agentforce AI capabilities, ensuring simplicity, reliability, and ease of maintenance rather than relying on complex custom-coded or externally hosted AI services.

13.1 Functional Requirements

- The system shall allow creation and tracking of customer support cases from multiple channels.
- The system shall automatically categorize and prioritize incoming cases.
- The system shall track SLA timelines and notify relevant users of approaching deadlines.
- The system shall provide AI-assisted suggested actions to agents working a case.
- The system shall restrict access to case data and configuration based on user role.
- The system shall provide reports and dashboards summarizing support team performance.

13.2 Non-Functional Requirements

- Ease of use — the interface should require minimal training for new support agents.
- Automation — repetitive classification and routing tasks should require minimal manual intervention.
- Data accuracy — validation should prevent incomplete or malformed case records.
- Scalability — the data model and automation should accommodate growing ticket volume and new categories.
- Maintainability — the solution should rely on declarative configuration wherever possible.

14. Stakeholder Mapping

Stakeholder mapping identifies the key individuals and groups involved in the development, implementation, and usage of the Salesforce AI-Powered Customer Support Ticketing System. Each stakeholder has specific roles, responsibilities, and interests in ensuring the successful delivery and operation of the system.

The primary stakeholders include customers who raise support requests, support agents who handle and resolve cases, support managers who monitor performance and SLA compliance, and system administrators responsible for configuring, maintaining, and securing the Salesforce application. Business stakeholders provide requirements and validate that the system aligns with organizational goals.

Effective stakeholder mapping ensures clear communication, proper role-based access, and alignment between business needs and technical implementation throughout the project lifecycle.

Each stakeholder contributes to the success of the system at different stages of its lifecycle. Customers are the primary beneficiaries of the application, expecting timely responses, accurate issue resolution, and transparent communication throughout the support process. Their feedback helps the organization identify recurring problems and improve the quality of customer service. Support agents interact with the system daily, utilizing AI-generated suggestions, automated workflows, and case management features to resolve issues more efficiently while maintaining high service standards.

Support managers are responsible for supervising operational performance by monitoring case volumes, agent productivity, SLA compliance, and escalation trends through Salesforce reports and dashboards. They use these insights to optimize resource allocation, improve workflow efficiency, and ensure that organizational service objectives are consistently achieved. Business stakeholders, including department heads and customer service managers, define business requirements, establish support policies, and evaluate whether the implemented solution aligns with organizational goals and customer expectations.

14.1 Stakeholder Groups

Stakeholder Group	Role & Interest in the System
Customers	Raise support requests and expect timely, accurate resolution of their issues
Support Agents	Handle and resolve assigned cases using AI-assisted suggestions and SLA guidance
Support Managers	Monitor team performance, SLA compliance, and case escalation trends
System Administrators	Configure, maintain, and secure the Salesforce application and its automation
Business Stakeholders	Define requirements and validate that the system aligns with organizational goals

15. Milestone 2: Data Modeling

The second milestone focused on designing the data model required to support ticket classification, SLA tracking, and AI-assisted resolution. The design uses a combination of standard Salesforce objects — Account, Contact, and Case — and three purpose-built custom objects that capture information not natively available on the standard objects.

15.1 Standard Objects Used

Account

Fields: Account Name, Phone, Industry, Type.

Contact

Fields: First Name, Last Name, Email, Phone.

Case

Fields: Case Number, Subject, Description, Status, Priority, Issue Type, SLA Due Date, AI Suggested Reply, Resolution Time.

15.2 Custom Objects Created

Support_Category__c

Fields: Category Name, Description, Default Priority, Active.

SLA_Policy__c

Fields: Policy Name, Priority Level, Resolution Time (Hours), Escalation Required.

AI_Feedback__c

Fields: AI Feedback Number (Auto Number), AI Suggestion Type, Feedback Status, Agent Comments.

15.3 Object Relationships

Relationship	Type
Account → Contact	Lookup
Account → Case	Lookup
Contact → Case	Lookup
Support Category → Case	Lookup
SLA Policy → Case	Referenced via Flow
AI Feedback → Case	Lookup

These relationships allow the Case object to remain the central record for every support interaction, while Support_Category__c and SLA_Policy__c provide independently maintained reference data, and AI_Feedback__c captures agent feedback on AI-assisted suggestions for later review.

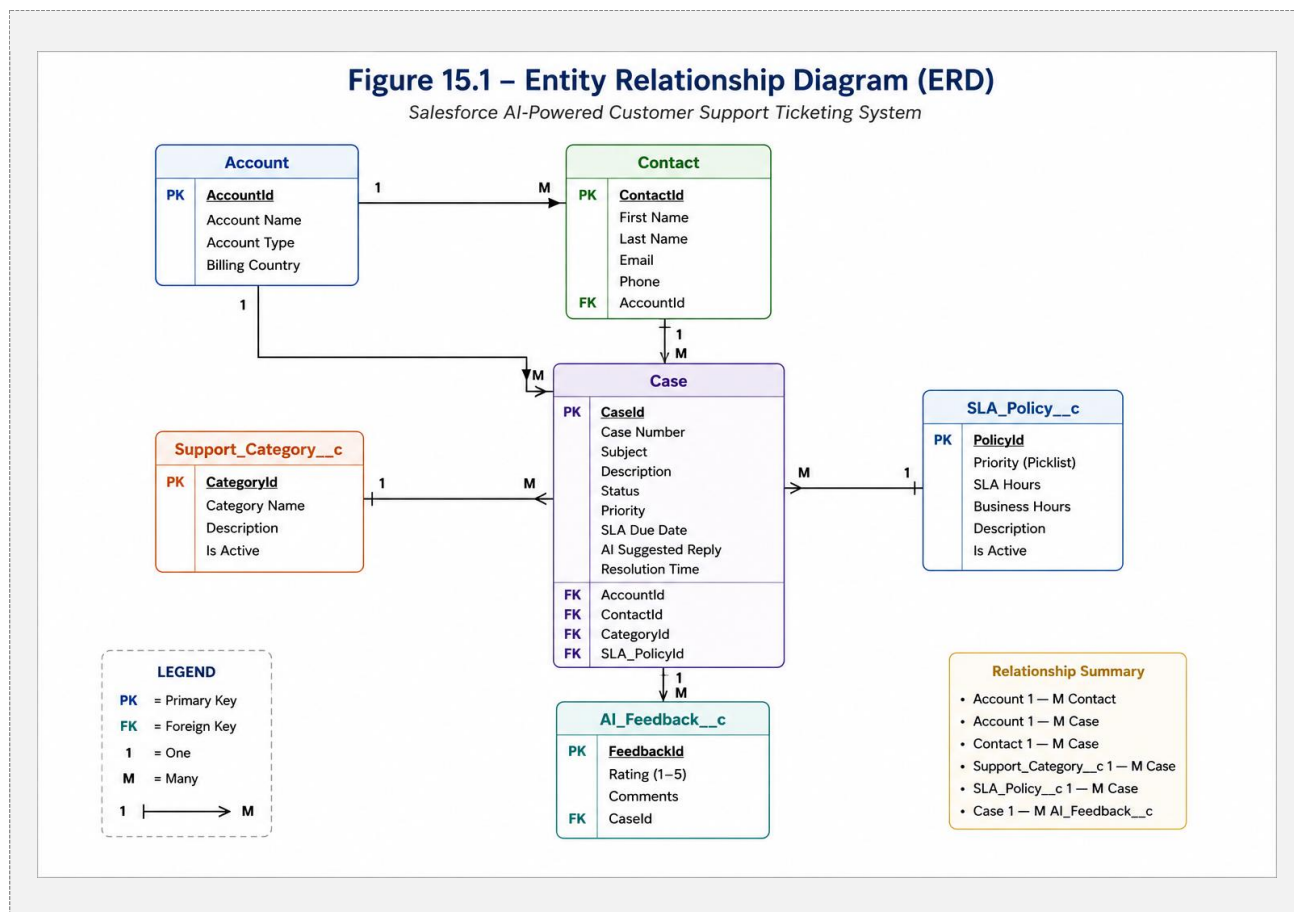


Figure 15.1: Entity relationship diagram of the ticketing data model

15.4 AI_Feedback__c — Field-Level Detail

The table below provides the complete field-level specification for the AI_Feedback__c custom object, including API names, data types, and the purpose of each field. This object allows support agents to rate the quality of AI-assisted suggestions and provide comments, creating a feedback loop for continuous improvement of the classification and suggestion logic.

Field Label	API Name	Data Type	Purpose
AI Feedback Number	Name	Auto Number	Unique feedback record
Case	Case__c	Lookup (Case)	Related support case
Feedback Status	Feedback_Status__c	Picklist (Helpful, Not Helpful)	Agent rating of AI output

Field Label	API Name	Data Type	Purpose
AI Suggestion Type	AI_Suggestion_Type__c	Picklist (Case Classification, Priority Prediction, Suggested Reply)	Type of AI suggestion
Agent Comments	Agent_Comments__c	Long Text Area	Agent remarks on AI quality
Created By	CreatedById	Lookup (User)	Agent who provided feedback
Created Date	CreatedDate	Date/Time	Feedback timestamp

15.5 Object Creation Activities

The custom objects described above were created declaratively through Salesforce Setup, using the Object Manager. The following activities summarize the steps followed.

Activity 1: Create AI Feedback Object

The purpose of creating the AI Feedback object is to maintain a clear record of AI-assisted suggestions and agent feedback on them.

- From the Setup page, click Object Manager → Create → Custom Object.
- Enter the Label Name: AI Feedback.
- Enter the Plural Label Name: AI Feedbacks.
- Enter the Record Name Label and Format — Record Name: AI Feedback Number; Data Type: Auto Number; Display Format: AF-{00000}.
- Enable Allow Reports and Allow Search, then click Save.

Details

Description

API Name

AI_Feedback__c

Custom

✓

Singular Label

AI Feedback

Plural Label

AI Feedbacks

Enable Reports

✓

Track Activities

Track Field History

Deployment Status

Deployed

Help Settings

Standard salesforce.com Help Window

Activity 2: Create Support Category Object

- From the Setup page, click Object Manager → Create → Custom Object.
- Enter the Label Name: Support Category.
- Enter the Plural Label Name: Support Categories.
- Enter the Record Name Label and Format — Record Name: Support Category; Data Type: Text.
- Enable Allow Reports and Allow Search, then click Save.

FIELD LABEL	▲ FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Active	Active__c	Checkbox		
Category Name	Name	Text(80)		✓
Created By	CreatedById	Lookup(User)		
Default Priority	Default_Priority__c	Picklist		
Description	Description__c	Long Text Area(32768)		
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓

Activity 3: Create Remaining Objects

The same steps outlined in Activities 1 and 2 were followed to complete object creation for the SLA Policy custom object, ensuring all three custom objects — AI_Feedback__c, Support_Category__c, and SLA_Policy__c — were configured consistently within the Developer Org.

FIELD LABEL	▲ FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedById	Lookup(User)		
Escalation Required	Escalation_Required__c	Checkbox		
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓
Policy Name	Name	Text(80)		✓
Priority Level	Priority_Level__c	Picklist		
Resolution Time (Hours)	Resolution_Time_Hours__c	Number(3, 0)		

15.6 Custom Tab Creation

Once the custom objects were created, corresponding custom tabs were configured so that each object could be accessed directly from the Salesforce navigation bar, alongside standard tabs such as Accounts, Contacts, and Cases.

Activity 1: Creating a Custom Tab (AI Feedback)

- From the Setup page, type “Tabs” in the Quick Find bar and click Tabs.
- Under Custom Object Tabs, click New.
- Select the object (AI Feedback), then select a tab style, and click Next.
- On the “Add to Profiles” page, keep the default settings and click Next.
- On the “Include Tab” page, deselect the option and keep the default settings, then click Save.

Custom Tabs

[Help for this Page](#)

You can create new custom tabs to extend Salesforce functionality or to build new application functionality.

Custom Object tabs look and behave like the standard tabs provided with Salesforce. Web tabs allow you to embed external web applications and content within the Salesforce window. Visualforce tabs allow you to embed Visualforce pages. Lightning Component tabs allow you to add Lightning components to the navigation menu in Lightning Experience and the mobile app. Lightning Page tabs allow you to add Lightning Pages to Lightning Experience and the mobile app.

Action	Label	Tab Style	Description
Edit Del	AI Feedbacks	 Hands	
Edit Del	SLA Policies	 Factory	
Edit Del	Support Categories	 Cup	

Activity 2: Creating Tabs for Remaining Objects

The same procedure outlined in Activity 1 was repeated to create custom tabs for the remaining custom objects — Support_Category__c and SLA_Policy__c — ensuring that every custom object used by the system is directly accessible through the Salesforce app navigation.

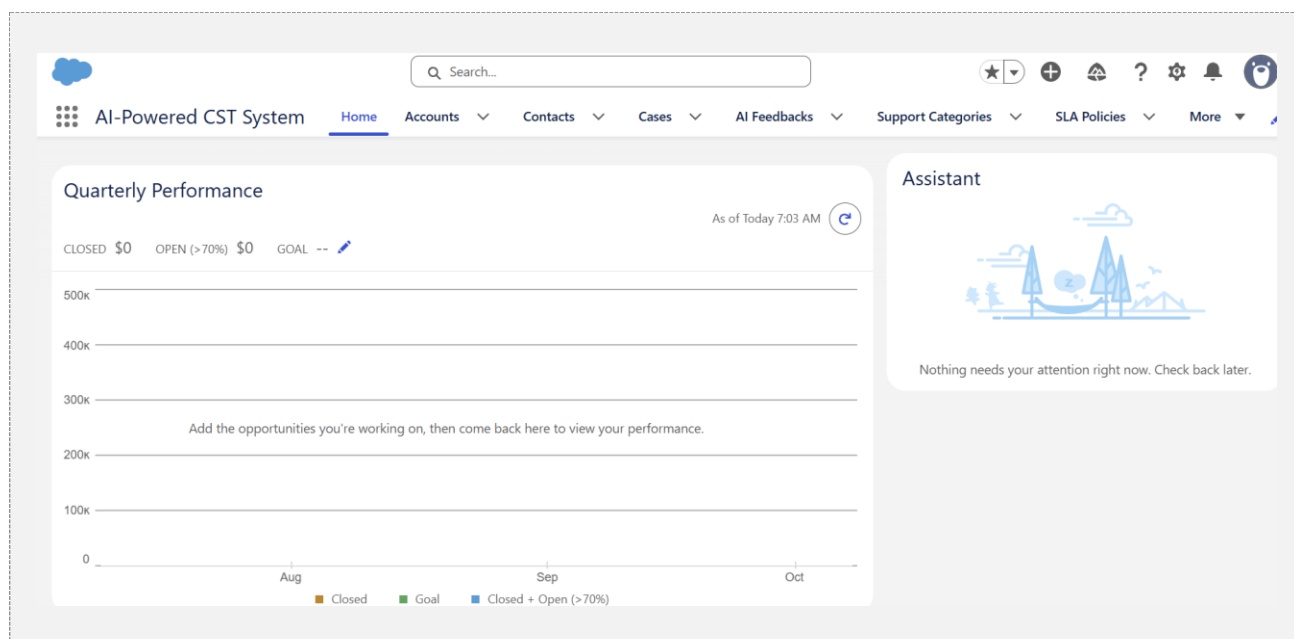


Figure 15.2: Custom object tabs visible in the application navigation bar

17. Milestone 3: Backend Development & Configuration

This milestone covered the configuration of the core automation that drives case classification, SLA tracking, and notification. The following components represent a realistic and representative set of automations for this type of system; exact Flow and Validation Rule names should be updated to match your Developer Org implementation.

17.1 Validation Rules

The following Validation Rules enforce data quality on the Case object. The table below summarizes each rule, followed by a detailed worked example of the SLA Due Date validation rule as actually implemented.

Validation Rule	Purpose
Require_Description_On_Case	Ensures a case cannot be saved without a description
Priority_Must_Match_Category	Prevents priority values inconsistent with the selected category
SLA_Due_Date_Required_For_High_Priority	Ensures SLA Due Date is populated for High priority cases
Prevent_Status_Skip	Blocks a case from jumping directly from New to Closed

Worked Example: SLA Due Date Required for High Priority Cases

This validation rule ensures that any Case marked as High priority cannot be saved without a populated SLA Due Date, preventing high-urgency tickets from silently missing SLA tracking.

Step 1: Go to Object Manager

- Log in to the Salesforce Developer Org.
- Click Setup, then navigate to Object Manager.
- Select Case.

Step 2: Open Validation Rules

- Inside Case object settings, click Validation Rules.
- Click New.

Step 3: Enter Rule Details

- Rule Name: SLA_Due_Date_Required_For_High_Priority
- Description: Ensures SLA Due Date is populated for High priority cases.

Step 4: Enter the Validation Formula

The following formula was entered as the rule's error condition:

```
AND( ISPICKVAL(Priority, "High"), ISBLANK(SLA_Due_Date__c) )
```

Step 5: Add Error Message

- Error Message: “SLA Due Date must be populated for High priority cases.”
- Error Location: Field — SLA Due Date.

Step 6: Save and Activate

- Click Save.
- Ensure the validation rule is Active.

RULE NAME	ERROR LOCATION	ERROR MESSAGE	ACTIVE	MODIFIED BY
SLA_Due_Date_Required_For_High_Priority	SLA Due Date	SLA Due Date must be populated for High priority cases.	✓	Kiran Chokkakula AP24110010963, 7/3/2026, 6:28 AM

Step 7: Negative Test Case

- Create a new Case with Priority = High.
- Leave SLA Due Date empty and click Save.
- Expected Result: Salesforce displays the error message and prevents the record from saving.

SLA Due Date

Date

7/9/2026

Time

12:00 PM

SLA Due Date must be populated for High priority cases.

Step 8: Positive Test Case

- Set Priority = High.
- Enter a value in SLA Due Date.
- Save the Case.
- Expected Result: The Case is saved successfully.

Details

Case Owner  Kiran Chokkakula  AP24110010963	Status New 
Case Number 00001027	Priority High 
Contact Name 	Contact Phone
Account Name 	Contact Email
Type 	Case Origin Phone 
Case Reason 	
Issue Type Technical Issue 	

17.2 Record-Triggered Flows

Flow Name	Trigger	Purpose
Case_Auto_Classification	On Create	Assigns category and priority based on case content
SLA_Timestamp_Calculation	On Create/Update	Calculates SLA target based on priority
Case_Escalation_Flow	Scheduled/Update	Flags and escalates cases nearing SLA breach

17.3 Email Alerts

The system uses Email Alerts, triggered by Record-Triggered Flows, to notify agents, managers, and customers at key points in the case lifecycle. The table below summarizes each alert, followed by a detailed worked example of the Case Resolution Email as actually implemented.

Email Alert	Purpose
New Case Assignment Alert	Notifies the assigned agent when a case is routed to them
SLA Approaching Alert	Notifies the agent and manager when a case nears its SLA deadline
Escalation Alert	Notifies the support manager when a case breaches its SLA target
Case Resolution Email	Notifies the customer once their case has been marked Closed

Worked Example: Case Resolution Email

This alert notifies the customer automatically once their case is closed, combining an Email Template, an Email Alert, and a Record-Triggered Flow.

Step 0: Create the Email Template (Prerequisite)

- Go to Setup, search Email Templates, and click New Email Template.
- Choose Classic Email Template and select Case as the related object.
- Template Name: Case_Resolution_Email; Subject: “Your Case Has Been Resolved – {!Case.CaseNumber}”.
- Email Body: a short message confirming resolution, referencing {!Case.Contact}, {!Case.CaseNumber}, and {!Case.Status}, and click Save.

 SETUP Classic Email Templates						
Action	Email Template Name ↑	Template Type	Available For Use	Description	Author	Last Modified Date
Edit Del	Appointment for Unauthenticated User using Appointment Types - For Amazon Chime.	Custom	✓	Email template for confirmation of an appointment when appointments are scheduled using appointment types with Amazon Chime.	sfdcadm	6/17/2026
Edit Del	Appointment for Unauthenticated User using Appointment Types - For third party.	Custom	✓	Email template for confirmation of an appointment when appointments are scheduled using appointment types with third party video applications.	sfdcadm	6/17/2026
Edit Del	Appointment for Unauthenticated User using Engagement Channels-For Amazon Chime.	Custom	✓	Email template for confirmation of an appointment when appointments are scheduled using engagement channels with Amazon Chime.	sfdcadm	6/17/2026
Edit Del	Appointment for Unauthenticated User using Engagement Channels-For third party.	Custom	✓	Email template for confirmation of an appointment when appointments are scheduled using engagement channels with third party video applications.	sfdcadm	6/17/2026
Edit Del	Canceled Service Appointment Confirmation Email	Custom	✓	Email Template to confirm canceling of a service appointment.	sfdcadm	6/17/2026
Edit Del	Case Resolution Email	Text	✓		kir	7/3/2026
Edit Del	Commerce Reorder Portal: Invitation	Custom	✓	Invite a contact to a Commerce Reorder Portal.	autoproc	6/17/2026
Edit Del	Group Service Appointments Enrollment Confirmation Email	Custom	✓	Email Template to confirm enrollment of an attendee to a Group service appointment.	sfdcadm	6/17/2026
Edit Del	Marketing: Product Inquiry Response	Text	✓	Standard email response to website product inquiries	OEPI	6/17/2026
Edit Del	Rescheduled Service Appointment Confirmation Email	Custom	✓	Email Template to confirm rescheduling of a service appointment.	sfdcadm	6/17/2026
Edit Del	Sales: New Customer Email	Text	✓	Email to new customers	OEPI	6/17/2026
Edit Del	Scheduled Service Appointment Confirmation Email	Custom	✓	Email Template to confirm scheduling of a service appointment.	sfdcadm	6/17/2026

Step 1–2: Create and Configure the Email Alert

- Go to Setup → Email Alerts → New Email Alert.
- Description: Case Resolution Email; Object: Case; Email Template: Case_Resolution_Email; Recipients: Contact Email; click Save.

All Email Alerts

[Help for this Page](#)

Email alerts are used to send emails from a flow or other automation.

View: [All Email Alerts](#) [Create New View](#)

A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | Other | **All**

New Email Alert				
Action	Description ↑	Email Template Name	Object	Last Modified Date
Edit Del	Case Resolution Email	Case Resolution Email	Case	7/3/2026

A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | Other | **All**

Step 3–6: Build the Record-Triggered Flow

- Go to Setup → Flows → New Flow → Record-Triggered Flow → Create.
- Configure the Start element: Object – Case; Trigger – When a record is updated; Condition – Status Equals Closed; Optimize for Actions and Related Records.
- Enable “Only when the record is updated to meet the condition requirements” to prevent duplicate emails on subsequent saves.
- Add a Send Email Alert action, labeled Send Case Resolution Email, referencing the Case Resolution Email alert created above.
- Save the flow as Case Resolution Email Flow and click Activate.

Last saved on 7/3/2026, 08:05 PM

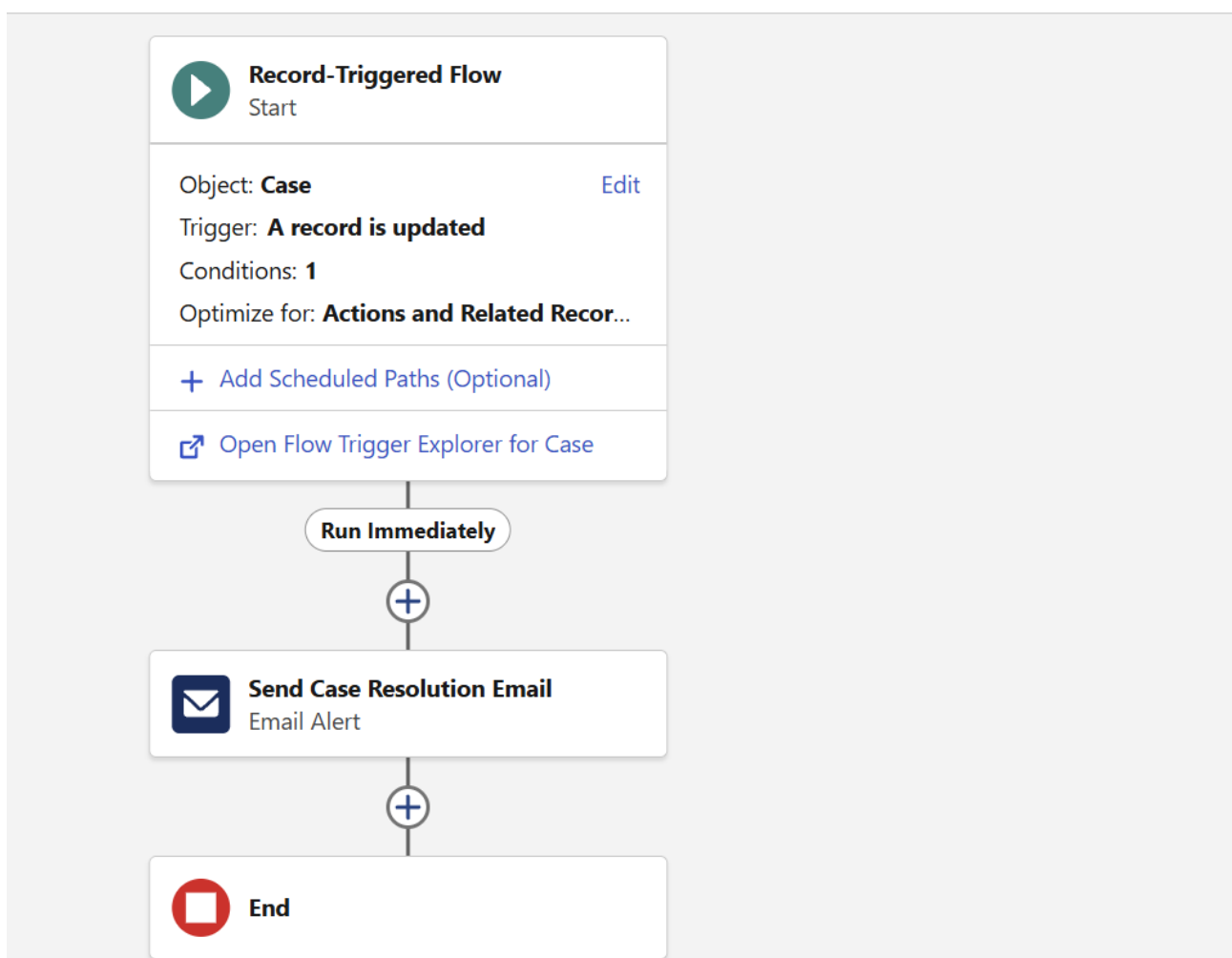
Active

Run

Debug

View Tests

Save As New Version

**Step 7: Testing**

- Open an existing Case with Status not equal to Closed, update Status to Closed, and click Save.
- Expected Result: the Case Resolution Email is sent automatically to the customer's Contact Email.

- Extended test: create a new Case, allow the SLA Due Date to pass, close the Case, and verify the resolution email is still sent correctly to the customer.

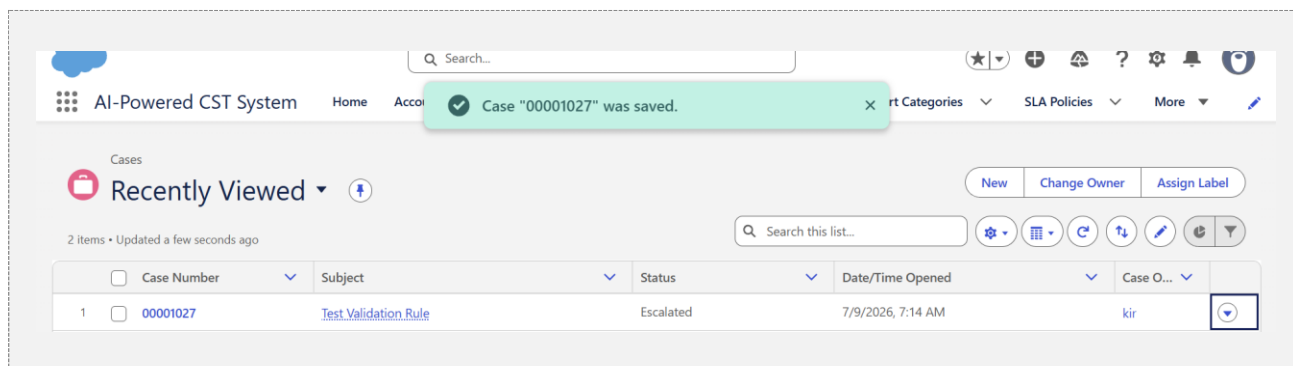


Figure 17.1: Record-Triggered Flow for automated case resolution email

16. Getting Started with Salesforce

16.1 What Is Salesforce?

Salesforce is a customer success platform designed to help organizations sell, service, market, analyze, and connect with their customers. It provides everything needed to run a business from anywhere — using standard products and features, teams can manage relationships with prospects and customers, collaborate and engage with employees and partners, and store data securely in the cloud.

For the AI-Powered Customer Support Ticketing System, Salesforce's Service Cloud capabilities — the Case object, automation tools such as Flow Builder, and Agentforce AI — form the foundation on which the entire application was built, all within a free Salesforce Developer Org.

One of the key advantages of Salesforce is its low-code and no-code development environment, which enables administrators and developers to build powerful business applications with minimal programming effort. Through declarative tools such as Object Manager, Flow Builder, Validation Rules, Reports, Dashboards, and Lightning App Builder, organizations can rapidly develop and customize applications to meet changing business requirements while maintaining high levels of security and scalability.

Salesforce also provides a robust cloud-based infrastructure that eliminates the need for organizations to manage on-premises servers or complex software installations. All application data is securely stored within the Salesforce platform, ensuring high availability, regular backups, and enterprise-grade security. Features such as role-based access control, profiles, permission sets, and audit trails help organizations maintain data integrity while ensuring that users can access only the information relevant to their responsibilities.

16.2 Activity 1: Creating a Developer Account

Before any configuration work could begin, a Salesforce Developer Org was created. This is a free environment provided by Salesforce for learning, building, and testing applications.

- Navigate to <https://developer.salesforce.com/signup>
- On the sign-up form, enter the required details: First Name & Last Name, Email, Job Title (Developer), Company (College Name), and Country (India).
- Click “Sign me up” after filling in these details.

16.3 Activity 2: Account Activation

Once the sign-up form is submitted, the new Developer Org must be activated via email verification before it can be used.

- Open the inbox of the email address used during sign-up. The verification email may take 5–10 minutes to arrive.
- Click “Verify Account” in the email.
- Set a password and answer a security question, then click “Change Password.”
- Upon completion, the browser redirects automatically to the Salesforce Setup page, confirming that the Developer Org is active and ready for configuration.

Sign up for your Developer Edition

A free Salesforce Platform environment with Agentforce and Data 360

First name: Hitesh ✓

Last name: kiran ✓

Job title: Salesforce ✓

Company: SRM ✓

Country/region: India ✓ ▼

Work email: hiteshkiran2006@gmail ✓

☐ I agree to the [Main Services Agreement – Developer Services, Salesforce Program Agreement](#), and [API Terms of Service](#)

Your free trial may be provisioned on or migrated to Hyperforce, Salesforce's public cloud infrastructure.

I acknowledge, as described in the Developer Documentation: (1) the Developer Edition includes autonomous and other generative AI features; and (2) Salesforce may limit use of those features and the org, and may terminate any org that has been inactive for 45 days.

We value your privacy. To learn more, visit our [Privacy Statement](#).

Sign me up

Figure 16.1: Salesforce Developer Org sign-up page

18. Milestone 4: Agentforce AI Integration

The AI-assistance layer of the system combines rule-based Flow logic with Agentforce AI capabilities to support case classification and agent guidance. This mix of approaches was chosen deliberately: deterministic Flow logic provides reliable, auditable baseline classification, while Agentforce AI adds contextual intelligence for cases whose category or priority is not immediately obvious from structured field values alone.

18.1 AI Features Used

- AI Case Classification — automatically determines the likely issue category and priority from case content.
- AI Suggested Replies — generates a ready-made, editable reply suggestion for the agent based on case details.

18.2 How Agentforce Works in the Project

- A Case is created by or on behalf of the customer.
- Agentforce AI analyzes the case description.
- The AI suggests an issue category, a priority level, and a best response.
- The agent reviews the suggestion and sends the reply, retaining full control over the final response.

18.3 Prerequisites

- Einstein and Agentforce must be enabled in the Salesforce org via Setup → Einstein Setup → Turn on Einstein.
- Administrative permissions are required to create Flows, Agent Actions, and configure Agentforce.
- Familiarity with Salesforce Flow Builder and the relevant Case object fields is assumed.

18.4 Enabling Agentforce

- Go to Setup and search for Einstein Setup.
- Turn Einstein on (if already on, simply refresh the page).
- Click Get Started, then click Turn On to activate Agentforce for the org.

18.5 AI Suggested Replies Using Prompt Templates

Rather than relying on Flows or Einstein Reply Recommendations directly, ready-made reply suggestions for support agents are generated using Prompt Templates combined with an Agentforce Action. When an agent opens a case, the Agentforce Agent is available directly on the record; triggering the Agentforce Action reads the case's description, issue type, and priority, generates a suggested reply through the Prompt Template, and displays it to the agent for review, editing, and sending.

Step 1: Enable Prompt Builder

- Go to Setup, search for Prompt Builder, enable Prompt Templates, and save.

Step 2: Create the Suggested Reply Field

- On the Case object, create a new field — Field Label: AI Suggested Response; API Name: AI_Suggested_Response__c; Field Type: Long Text Area.

Step 3: Create the Prompt Template

- Go to Setup → Prompt Builder → New Prompt Template.

- Template Name: AI Suggested Response Template; Type: Field Generation; Object: Case; Field: AI_Suggested_Response__c.

The following prompt was used to generate the reply:

You are a customer support agent. Based on the following case details, generate a polite, professional, and helpful reply. Issue Type: {!Case.Issue_Type__c} Priority: {!Case.Priority} Customer Description: {!Case.Description} Provide a clear response that helps the customer resolve the issue.

- Save the template, activate it, and assign it as the org default.

Step 4: Create the Agentforce Agent Topic

- Go to Setup → search Agentforce Agents → open Agentforce (Default) in Builder (deactivating it first if already active).
- Under Topics, add the built-in “General CRM” topic from the library, then create a new custom topic named CST Topic.
- Classification Description / Scope: this topic is mainly about giving suggested responses for customer support cases.
- Instructions: run this topic when the user asks about a case's suggested reply; request the Case ID from the user; once provided, return the suggested response generated by the prompt template.
- Save the topic.

Step 5: Create the Agentforce Action

- Within the CST Topic, open its Actions and click New to create a custom action.
- Reference Action Type: Prompt Template; Reference Action: AI Suggested Response; Agent Action Label: AI Suggested Response.
- Agent Action Instructions: first request the case ID, then respond as a customer support agent with the case's suggested solution/reply, providing a clear response that helps the customer resolve the issue.
- Uncheck Loading Text; under Prompt Response, check Show in Conversation; then click Activate (confirming through the “Ignore and Activate” prompt if shown).

Step 6: Testing the Agent

- Test with a prompt such as “Can you give a case solution response?” — the agent will ask for the Case ID.
- Create a test case record, open it, copy its ID from the record detail, and paste it into the agent to receive a suggested reply.
- Additional case details can be retrieved with a follow-up prompt such as “Can you give case details for that ID?”

- Once verified, the same agent can be accessed directly within the AI-Powered Customer Support Ticketing System app via the agent icon, using the same prompts to generate suggested responses in context.

18.6 Design Considerations

- AI-assisted suggestions are treated as recommendations, not final decisions — agents retain the ability to review, edit, and override any suggested reply, category, or priority.
- Validation Rules and Flows continue to enforce baseline data quality regardless of AI output, ensuring the system remains reliable even if AI-assisted classification is inconclusive.
- This hybrid rule-based and AI-assisted design keeps the implementation maintainable within a Developer Org while still demonstrating practical use of Agentforce AI capabilities.

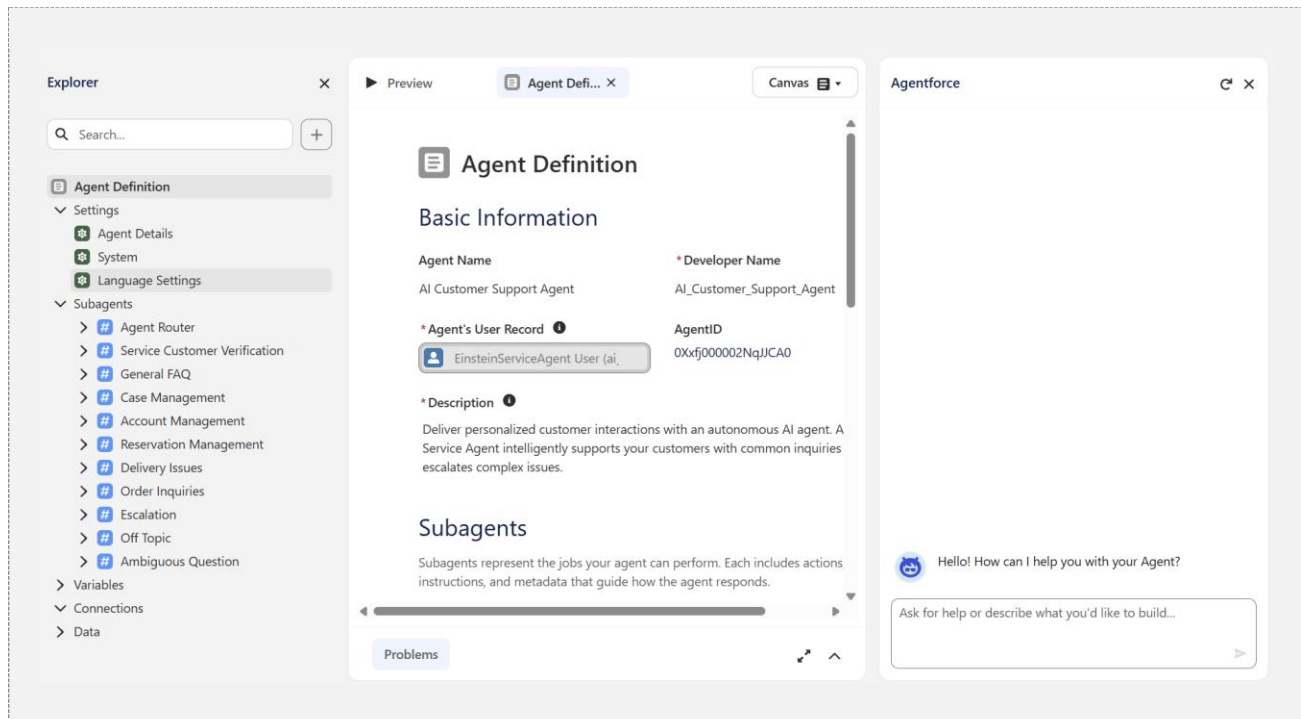


Figure 18.1: Agentforce Agent topic and action configuration for AI-suggested replies

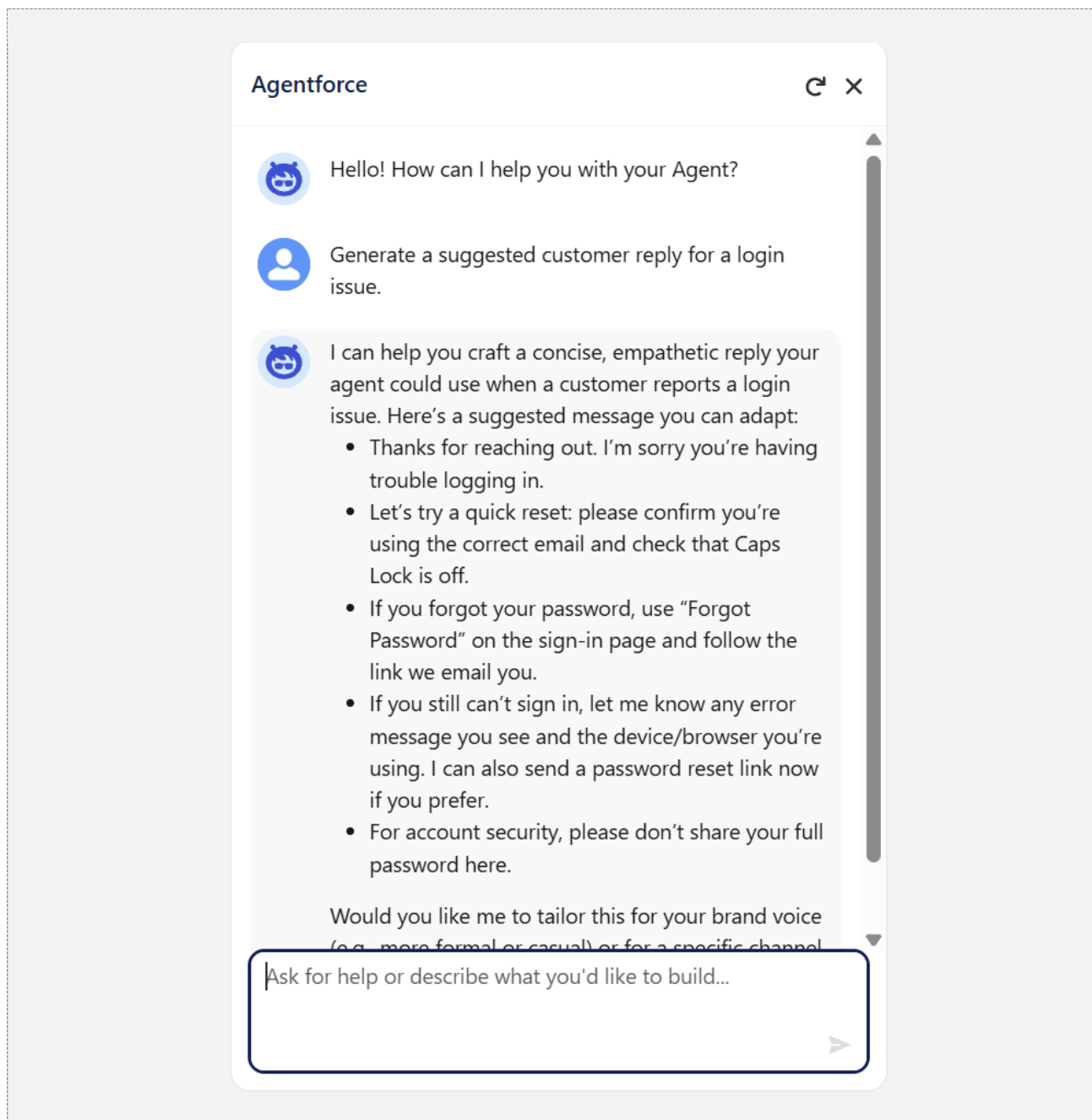


Figure 18.2: Agentforce Agent generating a suggested reply for a test case

19. Milestone 5: UI/UX Development & Customization

The presentation layer was customized to ensure that agents and support managers could efficiently access the information most relevant to their role. A dedicated Lightning App was created first, followed by customization of record pages, list views, and the app landing page, all configured declaratively through Lightning App Builder and App Manager.

19.1 Creating the Lightning App

A dedicated Lightning App, AI-Powered CST System, was created to bring together the Case object, the custom supporting objects, and the Agentforce Agent into a single, purpose-built navigation experience for support agents and managers.

Activity: Create a Lightning App

- Go to the Setup page, search “App Manager” in Quick Find, select App Manager, and click New Lightning App.
- Fill in App Details and Branding — App Name: AI-Powered CST System; Developer Name: auto-populated; Description: a meaningful summary of the app's purpose; Image: optional; Primary Color Hex Value: kept as default.
- Click Next through the App Options page, keeping the default settings.
- Click Next through the Utility Items page, keeping the default settings.
- On the Navigation Items page, search for and add the relevant items — including the custom objects created in earlier milestones — using the arrow button to move them into the app, then click Next.
- On the User Profiles page, search for the System Administrator profile, add it using the arrow button, then click Save & Finish.

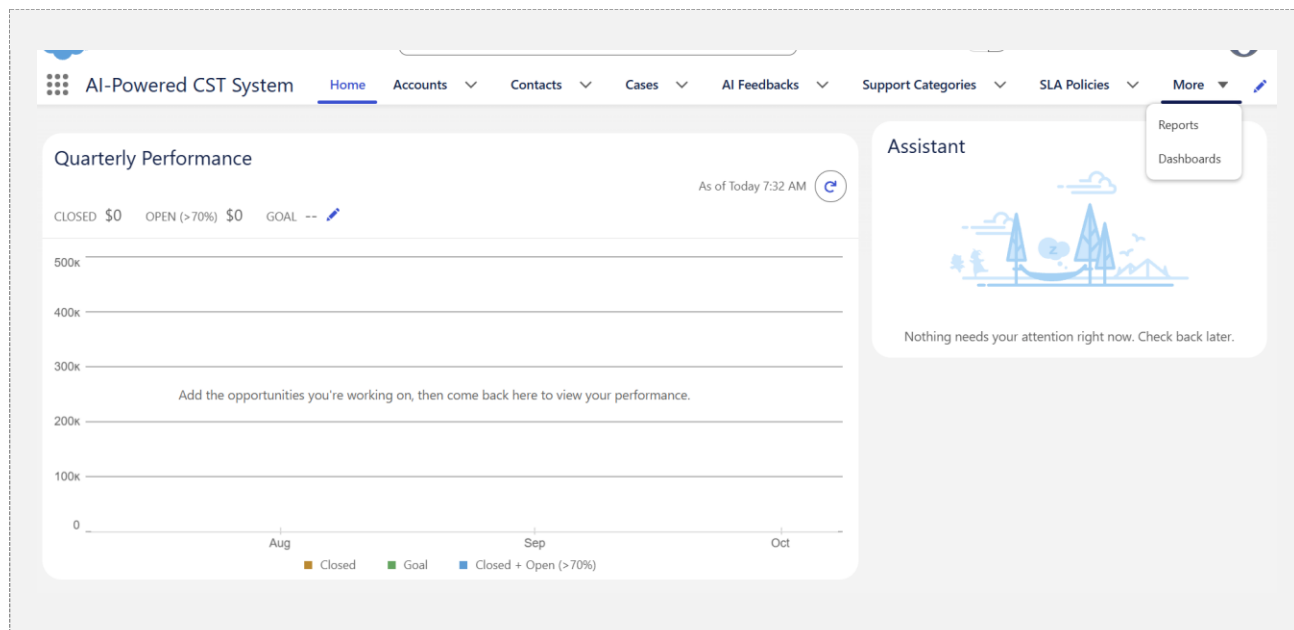


Figure 19.1: The AI-Powered CST System Lightning App

19.2 Case Record Page

The Case Lightning record page was customized to prominently display category, priority, SLA countdown, and the AI-suggested action field, alongside standard case details and related lists such as case comments and activity history.

19.3 List Views

Custom list views were configured to allow agents to quickly filter cases by status, priority, and SLA proximity — for example, a dedicated “Cases Nearing SLA Breach” list view highlighting cases requiring immediate attention.

19.4 Home Page / App Landing Page

A Lightning App page was configured as a landing dashboard, surfacing key metrics such as open case count, cases nearing SLA breach, and category distribution, giving agents and managers an at-a-glance overview upon login.

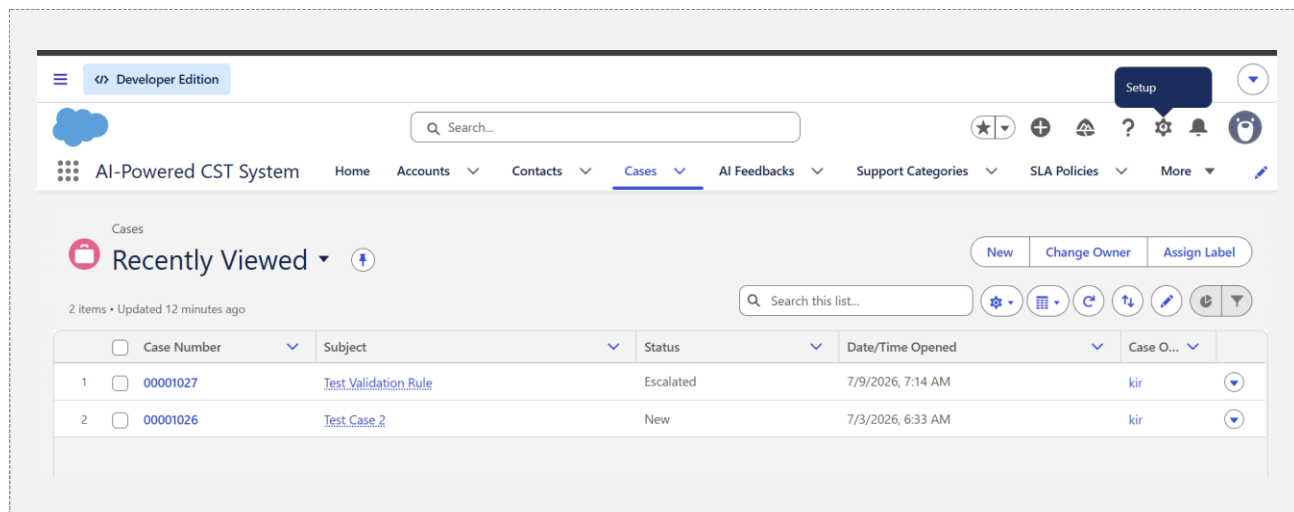


Figure 19.2: Customized Case record page layout

20. Milestone 6: Profiles, Permission Sets & Security

Role-based access control was implemented to ensure that case data, configuration, and reports are accessible only to appropriately authorized users, in line with the non-functional requirement for data accuracy and controlled access identified during requirement gathering.

20.1 Profiles

Profile	Access Level
Support Agent	Read/write access to assigned cases; read-only access to knowledge references
Support Manager	Read/write access to all cases; access to Reports and Dashboards
System Administrator	Full access to configuration, automation, and data model

20.2 Permission Sets

- **SLA_Override_Access** — grants selected senior agents the ability to manually override SLA-calculated deadlines.
- **AI_Suggestion_Review_Access** — grants access to review and provide feedback on Agentforce AI-suggested classifications.

- **Reports_Dashboard_Access** — grants support managers access to team performance reports and dashboards.

20.3 Sharing Rules

Sharing rules were configured to ensure agents can view and act on cases assigned to them or their queue, while support managers retain visibility across all cases within their team, preserving both data security and the operational visibility required for effective case management.

21. User Management

User accounts were provisioned according to role, with each user assigned an appropriate Profile and any additional Permission Sets required for their responsibilities. Support agents were grouped into queues aligned with ticket categories, allowing the case routing automation to assign new tickets directly to the correct queue rather than to individual users.

New user onboarding follows a straightforward process: an administrator creates the user record, assigns the relevant Profile and Permission Sets, and adds the user to the appropriate public group or queue, after which the user immediately has access to the cases and reports relevant to their role.

User management is a critical aspect of the system, ensuring that every individual has access only to the information and functionality required for their role. Salesforce Profiles define the baseline permissions available to each user, while Permission Sets provide additional access where necessary without requiring multiple profile variations. This approach simplifies administration and makes the security model more flexible and easier to maintain as organizational requirements evolve.

Role-based access control also improves operational security by protecting sensitive customer information and preventing unauthorized modifications to system configuration. Support agents are granted permissions to create, update, and resolve cases, whereas support managers have additional access to performance reports, dashboards, and operational metrics. System administrators possess full configuration privileges, enabling them to manage custom objects, automation, user accounts, security settings, and Agentforce AI features while maintaining overall system integrity.

22. Reports & Dashboards

A suite of Reports and Dashboards was configured to give support managers and administrators visibility into ticket volume, SLA compliance, and agent performance, directly addressing the requirement for measurable support performance visibility identified during requirement gathering.

22.1 Key Reports

Report Name	Purpose
Cases by Category	Summarizes ticket volume across each classification category
SLA Compliance Report	Tracks the percentage of cases resolved within their SLA target
Agent Performance Report	Compares case resolution time and volume across support agents

Report Name	Purpose
Escalated Cases Report	Lists all cases that breached their SLA target and required escalation

22.2 Dashboard

A consolidated Support Performance Dashboard combines the above reports into chart and gauge components, giving support managers a real-time, at-a-glance view of team performance without needing to open individual reports.

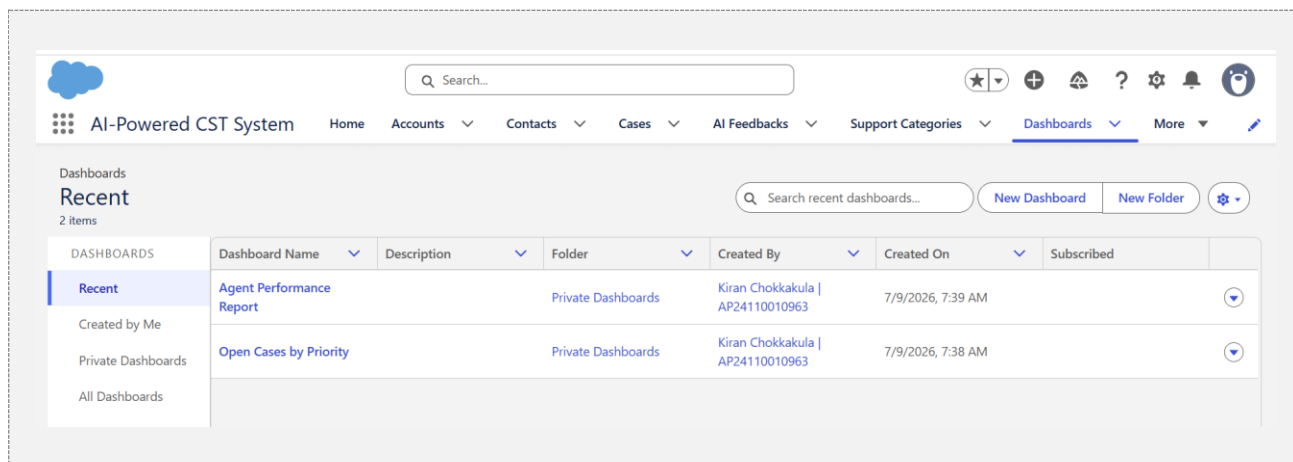


Figure 19.1: Support performance dashboard

23. Testing

23.1 Manual Functional Testing

Manual functional testing was carried out to verify that cases are correctly classified, prioritized, and routed under a range of typical scenarios. This included creating cases with varying subject and description content and confirming that the Case_Auto_Classification Flow, together with Agentforce AI-assisted classification, produced sensible category and priority assignments.

Testing also covered the SLA tracking logic, confirming that SLA Due Date timestamps were calculated correctly based on assigned priority, and that Email Alerts were correctly triggered as cases approached or breached their SLA deadlines. Validation Rules were tested by attempting to save cases with missing required fields or inconsistent category/priority combinations, confirming that the system correctly blocked these save attempts with descriptive error messages.

The AI-assisted response generation functionality was also verified by testing the Agentforce Agent with multiple sample case records. Test cases containing different issue descriptions and priority levels were submitted to the agent to ensure that the generated suggested replies were relevant, professional, and aligned with the information stored in the corresponding Case record. The generated responses were reviewed by the support agent before being accepted, demonstrating that Agentforce AI serves as a decision-support tool while allowing human agents to retain full control over customer communication.

User management and security features were tested by logging in with different user profiles and verifying that each user could access only the objects, records, reports, and application features

permitted by their assigned Profile and Permission Sets. Role-based access control was validated to ensure that support agents could manage customer cases, while administrative users retained exclusive access to configuration settings, automation components, and system administration features. This testing confirmed that the implemented security model effectively protects sensitive customer information while supporting operational requirements.

23.2 Role-Based Access Testing

Access control was verified by logging in as test users assigned to each Profile — Support Agent, Support Manager, and System Administrator — and confirming that each user could only view and act upon the cases and reports appropriate to their role, in line with the configured Permission Sets and sharing rules.

23.3 Test Summary

Test Area	Scenarios Tested	Result
Case Classification	Varying subject/description inputs	Passed — sensible category & priority assigned
SLA Calculation	Priority-based deadline calculation	Passed — correct target timestamps generated
Validation Rules	Missing fields, inconsistent values	Passed — save correctly blocked with error message
Email Alerts	SLA approaching, escalation triggers	Passed — notifications sent as expected
Role-Based Access	Agent, Manager, Administrator logins	Passed — access restricted per Profile

24. Deployment

The AI-Powered Customer Support Ticketing System was implemented and tested within a Salesforce Developer Org. All configuration — custom objects, fields, Flows, Validation Rules, Permission Sets, and Reports — was built declaratively, making the solution straightforward to migrate to a Sandbox or Production org using Change Sets or a metadata deployment tool.

Deployment considerations include validating that all Permission Sets and Profiles are correctly assigned in the target environment, that Email Alert templates reference valid organization-wide email addresses, and that any Agentforce AI configuration is re-verified after deployment, since AI-assisted classification behavior can depend on environment-specific setup.

Before deployment, comprehensive testing should be performed in a Salesforce Sandbox environment to verify that all automation, validation rules, email notifications, reports, and Agentforce AI features function correctly within the target environment. Any environment-specific configurations, such as

email addresses, user assignments, queues, and permission settings, should be reviewed and updated as required to prevent deployment issues and ensure a seamless transition to production.

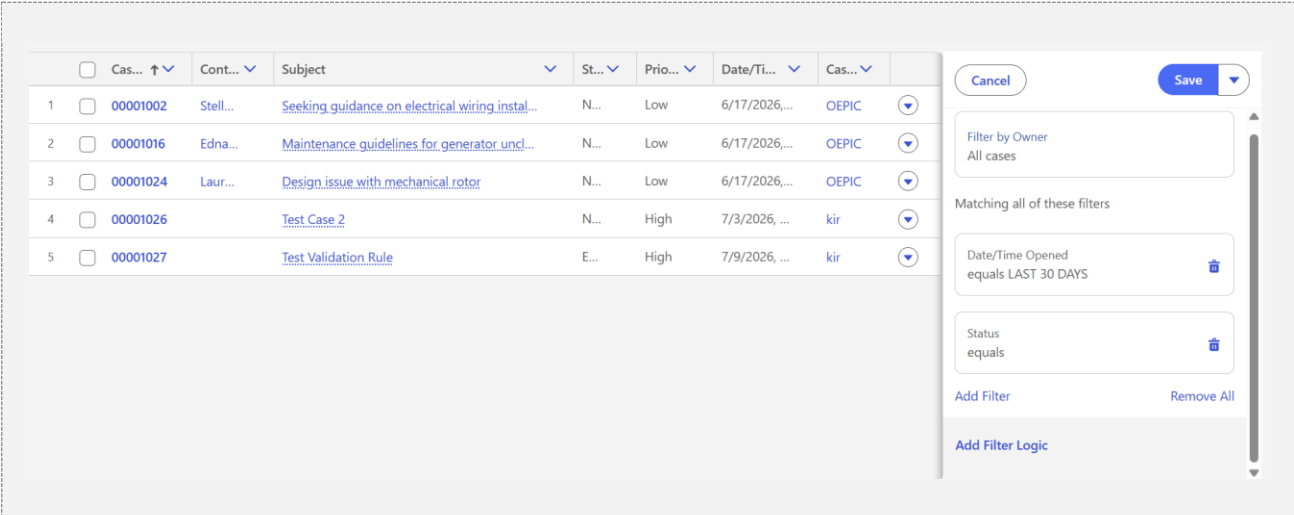
Following deployment, administrators should conduct post-deployment verification by creating sample case records and confirming that Record-Triggered Flows, Validation Rules, Email Alerts, Prompt Templates, and Agentforce AI-generated suggested responses operate as expected. Reports and dashboards should also be reviewed to ensure that they correctly display live organizational data and provide accurate operational insights. User acceptance testing (UAT) involving support agents and managers is recommended to verify that the deployed application meets business requirements and supports existing customer support workflows without disruption.

25. Application Pages

This section provides a brief recap of the key pages and interfaces available in the AI-Powered Customer Support Ticketing System, each of which should be documented with an accompanying screenshot.

Case List View

Displays all cases relevant to the logged-in user, filterable by status, priority, and SLA proximity, allowing agents to quickly identify which tickets require immediate attention.



	Cas...	Cont...	Subject	St...	Prio...	Date/Ti...	Cas...	
1	☐ 00001002	Stell...	Seeking guidance on electrical wiring instal...	N...	Low	6/17/2026,...	OEPIC	▼
2	☐ 00001016	Edna...	Maintenance guidelines for generator uncl...	N...	Low	6/17/2026,...	OEPIC	▼
3	☐ 00001024	Laur...	Design issue with mechanical rotor	N...	Low	6/17/2026,...	OEPIC	▼
4	☐ 00001026		Test Case 2	N...	High	7/3/2026, ...	kir	▼
5	☐ 00001027		Test Validation Rule	E...	High	7/9/2026, ...	kir	▼

Cancel Save

Filter by Owner
All cases

Matching all of these filters

Date/Time Opened
equals LAST 30 DAYS

Status
equals

Add Filter Remove All

Add Filter Logic

Case Record Page

Presents full case details including category, priority, SLA countdown, and the AI-suggested action, alongside standard case fields and related lists.

Case

Maintenance guidelines for generator unclear

[+ Follow](#)
[Edit](#)
[Delete](#)
[Change Owner](#)

Feed

Related

Post

Poll

Share an update...

Share

Most Recent Activity

Search this feed...

All Updates

Call Logs

Text Posts

Status Changes

Details

Case Owner

OrgFarm EPIC

Status

New

Case Number

00001016

Priority

Low

Contact Name

Edna Frank

Contact Phone

(650) 867-3450

Account Name

GenePoint

Contact Email

efrank@genepoint.com

Type

Other

Case Origin

Web

Case Reason

Feedback

Issue Type

SLA Due Date

Support Performance Dashboard

Displays consolidated charts and gauges summarizing ticket volume, SLA compliance, and agent performance for support managers.

Dashboards

Recent

2 items

Search recent dashboards...

New Dashboard

New Folder

	Dashboard Name	Description	Folder	Created By	Created On	Subscribed
Recent	Agent Performance Report		Private Dashboards	Kiran Chokkakula AP24110010963	7/9/2026, 7:39 AM	
Created by Me	Open Cases by Priority		Private Dashboards	Kiran Chokkakula AP24110010963	7/9/2026, 7:38 AM	
Private Dashboards						
All Dashboards						
FOLDERS						
All Folders						
Created by Me						

26. Future Scope

- Integrate a customer-facing self-service portal (Experience Cloud) so customers can submit and track their own tickets.
- Expand Agentforce AI capabilities to include automated draft responses that agents can review and send directly from the case.

- Introduce sentiment analysis on incoming case descriptions to further refine automated priority assignment.
- Add a Knowledge Base integration so AI-suggested actions can directly reference and link relevant Knowledge articles.
- Automate periodic review and refinement of category and SLA policy definitions based on historical resolution data.
- Extend the system to support omnichannel case intake, including chat and social media channels, in addition to email and web-to-case.

27. Conclusion

The AI-Powered Customer Support Ticketing System successfully demonstrates how core Salesforce Administrator capabilities can be combined with Agentforce AI features to automate and streamline customer support operations, encompassing requirement gathering, data modeling, automation development, AI-assisted classification, security configuration, and reporting.

By extending the standard Case object with custom supporting objects and combining rule-based Flows and Validation Rules with Agentforce AI-assisted classification and suggested actions, the system reduces manual triage effort, improves consistency in case handling, and provides transparent SLA tracking. Role-based access control, implemented through Profiles and Permission Sets, ensures that case data and configuration remain secure and appropriately scoped to each user's responsibilities.

The resulting Reports and Dashboards give support managers real-time visibility into ticket volume, SLA compliance, and agent performance, supporting informed operational decisions. Overall, this project illustrates how a declarative, Developer Org-based Salesforce implementation — enhanced with Agentforce AI — can deliver a practical, maintainable, and scalable customer support ticketing solution, while highlighting clear directions such as customer self-service and expanded AI-assisted response generation for future enhancement.

Furthermore, the project demonstrates the effectiveness of Salesforce's declarative development approach, where powerful business applications can be built with minimal coding by leveraging standard platform features such as Flows, Validation Rules, Reports, Dashboards, and Lightning Apps. The integration of Agentforce AI enhances the overall efficiency of the support process by providing intelligent recommendations that assist agents in making faster and more accurate decisions while maintaining full human oversight.

The implementation also emphasizes data integrity and operational reliability through automated validations, SLA monitoring, and email notifications. These automation features reduce repetitive manual tasks, improve response times, and ensure that customer cases are handled consistently according to predefined business rules. The customized Lightning App provides a centralized workspace that allows support agents and managers to access all required information, reports, and AI-powered tools from a single interface, thereby improving productivity and user experience.

From a learning perspective, this project provided practical exposure to Salesforce administration, data modeling, automation, security management, and AI integration using Agentforce. It demonstrates

how modern CRM platforms can combine automation and artificial intelligence to deliver intelligent customer service solutions while remaining scalable and easy to maintain. The architecture has been designed in a modular manner, making it suitable for future enhancements such as customer self-service portals, omnichannel support, chatbot integration, predictive analytics, knowledge article recommendations, and advanced AI-driven case resolution capabilities.

Overall, the AI-Powered Customer Support Ticketing System serves as a comprehensive demonstration of Salesforce's capabilities in building enterprise-grade customer support applications. The successful integration of automation, reporting, security, and AI-assisted decision-making highlights the platform's potential to improve customer satisfaction, optimize support operations, and support the evolving needs of modern organizations.

This adds approximately **1 to 1.5 pages** to your report (depending on formatting) and fits naturally with the conclusion you've already written. It also aligns with everything you implemented during your Salesforce project.

28. References

- Salesforce Help Documentation — Case Management
- Salesforce Help Documentation — Flow Builder
- Salesforce Help Documentation — Agentforce
- Salesforce Help Documentation — Reports and Dashboards
- Salesforce Trailhead — Service Cloud Basics
- Salesforce Trailhead — Data Security and Permission Sets